

Solutions to CMI Mock Test (Part B)

Question 1

(a) Nature of the discontinuity at $x = 0$

For $x \neq 0$, $f(x) = x \sin(1/x)$. Since $-1 \leq \sin(1/x) \leq 1$, multiplying by $|x|$ gives

$$-|x| \leq x \sin \frac{1}{x} \leq |x|.$$

Both bounds tend to 0 as $x \rightarrow 0$, so by the Squeeze Theorem

$$\lim_{x \rightarrow 0} f(x) = 0.$$

The limit exists and is finite, but $f(0) = 1 \neq 0$. Hence f has a **removable discontinuity** at $x = 0$.

(b) Failure of the Intermediate Value Property on $[-1, 1]$

Consider the sub-interval $[0, 1/\pi]$. At its endpoints,

$$f(0) = 1, \quad f\left(\frac{1}{\pi}\right) = \frac{1}{\pi} \sin \pi = 0.$$

Take $k = \frac{1}{2}$, so $0 < k < 1$. If f had the IVP on $[-1, 1]$, some $c \in (0, 1/\pi)$ would satisfy $f(c) = \frac{1}{2}$. But for every $x \in (0, 1/\pi)$,

$$|f(x)| = \left|x \sin \frac{1}{x}\right| \leq |x| < \frac{1}{\pi} < \frac{1}{2}.$$

So $f(x) < \frac{1}{2}$ strictly throughout the open interval, and the value $\frac{1}{2}$ is never attained. Hence f does not satisfy the IVP on $[-1, 1]$.

(c) A discontinuity satisfying the IVP must be essential

Suppose g has a discontinuity at $x = a$ and satisfies the IVP on every interval containing a . We show the discontinuity cannot be removable or a jump.

Case 1: Removable discontinuity. Both one-sided limits equal some common value L , with $L \neq g(a)$. Assume $g(a) > L$ (the other case is symmetric). Pick k with $L < k < g(a)$. Since $\lim_{x \rightarrow a} g(x) = L$, there is $\delta > 0$ such that $g(x) < k$ for every $x \in (a - \delta, a + \delta) \setminus \{a\}$. Fix $x_1 \in (a - \delta, a)$. Then

$$g(x_1) < k < g(a),$$

so by the IVP some $c \in (x_1, a)$ gives $g(c) = k$. This contradicts $g(x) < k$ on the punctured neighbourhood.

Case 2: Jump discontinuity. Let $L^- = \lim_{x \rightarrow a^-} g(x)$ and $L^+ = \lim_{x \rightarrow a^+} g(x)$ with $L^- \neq L^+$; assume $L^- < L^+$. Pick $k \in (L^-, L^+)$ with $k \neq g(a)$. By definition of one-sided limits, there is $\delta > 0$ such that

$$g(x) < k \text{ for } x \in (a - \delta, a), \quad g(x) > k \text{ for } x \in (a, a + \delta).$$

Choose $x_1 \in (a - \delta, a)$ and $x_2 \in (a, a + \delta)$. Then $g(x_1) < k < g(x_2)$, but g takes no value equal to k on (x_1, x_2) : on (x_1, a) we have $g < k$, on (a, x_2) we have $g > k$, and $g(a) \neq k$ by choice. This contradicts the IVP on $[x_1, x_2]$.

Both non-essential cases fail, so the discontinuity at a must be essential. ■

Question 2

Construction of the axis of a parabola

The construction rests on the diameter property: *the midpoints of any family of parallel chords of a parabola lie on a line parallel to its axis*. The unique diameter which is perpendicular to its own associated family of chords is the axis itself.

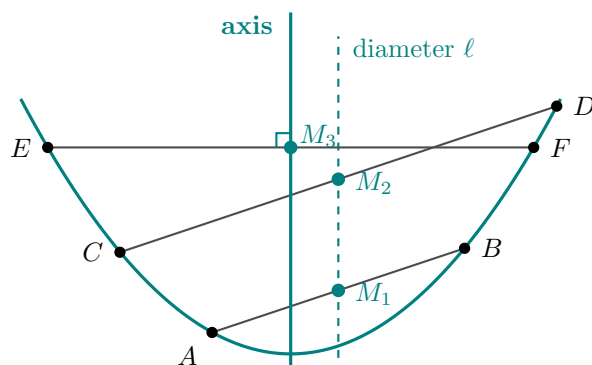
Step 1. Draw two parallel chords. With the straightedge draw any secant cutting the parabola at A, B . Draw a second line parallel to AB cutting the parabola at C, D .

Step 2. Bisect the chords. With compass and straightedge, construct the perpendicular bisectors of AB and CD ; these meet AB, CD at their midpoints M_1, M_2 .

Step 3. Construct a diameter. The line $\ell = M_1M_2$ is a diameter, hence parallel to the axis.

Step 4. Draw a chord perpendicular to the diameter. Through any convenient point of the parabola, construct a line perpendicular to ℓ ; it meets the parabola at E, F . The chord EF is perpendicular to ℓ and so to the axis.

Step 5. Construct the axis. Let M_3 be the midpoint of EF . The line through M_3 parallel to ℓ is a diameter bisecting a chord perpendicular to itself. By the reflective symmetry of the parabola, this line is the axis of symmetry.



*The chords $AB \parallel CD$ give midpoints M_1, M_2 ; the line M_1M_2 is a diameter ℓ .
The chord $EF \perp \ell$ has midpoint M_3 ; the line through M_3 parallel to ℓ is the axis.*

Remark. The other axis of symmetry (the tangent at the vertex) is simply the perpendicular to the axis already constructed, drawn through the vertex of the parabola.

Question 3

(a) Counter-example for arbitrary matrices

Expanding directly, $(A+B)^2 = A^2 + AB + BA + B^2$, so the identity $(A+B)^2 = A^2 + 2AB + B^2$ is equivalent to $AB = BA$. Take

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

Then

$$A+B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad (A+B)^2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

On the other hand, $A^2 = B^2 = 0$, while

$$AB = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad A^2 + 2AB + B^2 = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}.$$

The two matrices differ, so the binomial identity fails for this pair.

(b) Proof when $AB = BA$

Assume $AB = BA$. We first show that powers of A and B commute.

Lemma 1. $A^k B = B A^k$ for every integer $k \geq 1$.

Induction on k . The case $k = 1$ is given. Assuming $A^m B = B A^m$,

$$A^{m+1} B = A(A^m B) = A(B A^m) = (AB) A^m = (BA) A^m = B A^{m+1}.$$

Lemma 2. $A^k B^j = B^j A^k$ for all integers $j, k \geq 1$.

Fix k and induct on j . The case $j = 1$ is Lemma 1. Assuming $A^k B^n = B^n A^k$,

$$A^k B^{n+1} = (A^k B^n) B = (B^n A^k) B = B^n (A^k B) = B^n (B A^k) = B^{n+1} A^k,$$

using Lemma 1 at the penultimate step.

Main induction. We prove

$$(A+B)^n = \sum_{r=0}^n \binom{n}{r} A^{n-r} B^r$$

by induction on n . The case $n = 1$ is immediate. Assume the formula for $n = m$:

$$(A+B)^{m+1} = (A+B) \sum_{r=0}^m \binom{m}{r} A^{m-r} B^r = \sum_{r=0}^m \binom{m}{r} A^{m+1-r} B^r + \sum_{r=0}^m \binom{m}{r} B A^{m-r} B^r.$$

By Lemma 2, $BA^{m-r} = A^{m-r}B$, so the second sum becomes $\sum_{r=0}^m \binom{m}{r} A^{m-r} B^{r+1}$. Shifting the index $r \rightarrow r-1$ and combining with the first sum,

$$(A+B)^{m+1} = A^{m+1} + \sum_{r=1}^m \left[\binom{m}{r} + \binom{m}{r-1} \right] A^{m+1-r} B^r + B^{m+1}.$$

Pascal's identity $\binom{m}{r} + \binom{m}{r-1} = \binom{m+1}{r}$ gives exactly the expansion for $n = m+1$. ■

Question 4

(a) Values of A_1, A_2, A_3

- A_1 : a 2×1 strip admits only a single vertical domino. So $A_1 = 1$.
- A_2 : a 2×2 square can be tiled by two vertical dominoes, two horizontal dominoes, or one 2×2 square tile. So $A_2 = 3$.
- A_3 : splitting on the rightmost column gives one vertical domino (leaving a 2×2 , i.e. 3 ways), two horizontal dominoes (leaving a 2×1 , i.e. 1 way), or a square tile (leaving a 2×1 , i.e. 1 way). So $A_3 = 3 + 1 + 1 = 5$.

(b) Recurrence

Fix $n \geq 3$ and classify tilings of the $2 \times n$ grid by how the rightmost column(s) are covered. The cases are mutually exclusive and exhaustive:

- (i) A vertical domino fills the last column; the remaining $2 \times (n - 1)$ grid is tiled in A_{n-1} ways.
- (ii) Two stacked horizontal dominoes fill the last two columns; the remaining $2 \times (n - 2)$ grid is tiled in A_{n-2} ways.
- (iii) A single 2×2 square tile fills the last two columns; again A_{n-2} ways remain.

Summing,

$$\boxed{A_n = A_{n-1} + 2A_{n-2}} \quad (n \geq 3).$$

(c) Closed form

We prove $A_n = \frac{2^{n+1} + (-1)^n}{3}$ for every $n \geq 1$ by strong induction.

Base cases. $\frac{2^2 + (-1)^1}{3} = \frac{3}{3} = 1 = A_1$ and $\frac{2^3 + (-1)^2}{3} = \frac{9}{3} = 3 = A_2$.

Inductive step. Assume the formula holds for all indices less than n . Then

$$\begin{aligned} A_n &= A_{n-1} + 2A_{n-2} \\ &= \frac{2^n + (-1)^{n-1}}{3} + 2 \cdot \frac{2^{n-1} + (-1)^{n-2}}{3} \\ &= \frac{2^n - (-1)^n + 2^n + 2(-1)^n}{3} = \frac{2 \cdot 2^n + (-1)^n}{3} = \frac{2^{n+1} + (-1)^n}{3}. \end{aligned}$$

This closes the induction. ■

Question 5

(a) The bound $|f(x)| \leq \sqrt{x}$

Since f is continuously differentiable and $f(0) = 0$, the Fundamental Theorem of Calculus gives

$$f(x) = \int_0^x f'(t) dt.$$

Applying the Cauchy–Schwarz inequality to the functions 1 and f' on $[0, x]$,

$$|f(x)|^2 = \left(\int_0^x 1 \cdot f'(t) dt \right)^2 \leq \left(\int_0^x 1^2 dt \right) \left(\int_0^x (f'(t))^2 dt \right) = x \int_0^x (f'(t))^2 dt.$$

Because $(f'(t))^2 \geq 0$ and $x \leq 1$,

$$\int_0^x (f'(t))^2 dt \leq \int_0^1 (f'(t))^2 dt = 1.$$

Therefore $|f(x)|^2 \leq x$, and taking square roots, $|f(x)| \leq \sqrt{x}$ for every $x \in [0, 1]$.

(b) The integral bound

From part (a), $f(x) \leq |f(x)| \leq \sqrt{x}$ on $[0, 1]$. Integrating,

$$\int_0^1 f(x) dx \leq \int_0^1 \sqrt{x} dx = \left[\frac{2}{3} x^{3/2} \right]_0^1 = \frac{2}{3}.$$

(c) When equality holds

Equality in the Cauchy–Schwarz step of part (a) forces f' to be proportional to the constant 1 on each interval $[0, x]$, and equality in the integrated bound forces $f(x) = \sqrt{x}$ almost everywhere. Test this candidate against the premise: $f(x) = \sqrt{x}$ gives

$$f'(x) = \frac{1}{2\sqrt{x}},$$

which is undefined at $x = 0$, so f is not continuously differentiable on $[0, 1]$. Moreover,

$$\int_0^1 (f'(x))^2 dx = \int_0^1 \frac{1}{4x} dx = +\infty,$$

violating $\int_0^1 (f')^2 = 1$. The only candidate for equality fails both hypotheses, so no function f satisfying the stated conditions attains equality.

Question 6

Throughout, $p > 3$ is prime. Write S_k for the k -th elementary symmetric sum of $1, 2, \dots, p-1$.

(a) p divides the sum of squares

By the standard formula,

$$\sum_{k=1}^{p-1} k^2 = \frac{(p-1)p(2p-1)}{6} = p \cdot \frac{(p-1)(2p-1)}{6}.$$

Since $p > 3$ is prime, $\gcd(p, 6) = 1$, so 6 divides $(p-1)(2p-1)$ and the bracketed quantity is an integer. The sum is therefore an integer multiple of p .

(b) p divides S_1 and S_2

Consider the polynomial

$$P(x) = (x-1)(x-2)\cdots(x-(p-1)),$$

whose roots are $1, 2, \dots, p-1$. By Vieta,

$$S_1 = \sum_{k=1}^{p-1} k = \frac{(p-1)p}{2} = p \cdot \frac{p-1}{2}.$$

Since p is odd, $(p-1)/2$ is an integer, so $p \mid S_1$. Newton's identity relates S_1, S_2 and the power sum $P_2 = \sum k^2$:

$$2S_2 = S_1^2 - P_2.$$

By part (a), $p \mid P_2$, and from above $p \mid S_1$, so $p \mid 2S_2$. As $\gcd(2, p) = 1$, it follows that $p \mid S_2$.

(c) A is divisible by p and by p^2

Combining terms over a common denominator,

$$H_{p-1} = \sum_{k=1}^{p-1} \frac{1}{k} = \frac{S_{p-2}}{(p-1)!} = \frac{A}{B},$$

where S_{p-2} is the elementary symmetric sum of $1, 2, \dots, p-1$ taken $p-2$ at a time (i.e. the sum of products omitting one factor). Since $(p-1)!$ is coprime to p , divisibility of A by a power of p is equivalent to divisibility of S_{p-2} by the same power.

Divisibility by p . Expanding,

$$P(x) = x^{p-1} - S_1x^{p-2} + S_2x^{p-3} - \cdots + (-1)^{p-1}S_{p-1}.$$

By Fermat's Little Theorem, $k^{p-1} \equiv 1 \pmod{p}$ for $k = 1, 2, \dots, p-1$. Hence each root of P is also a root of $x^{p-1} - 1$ modulo p , and since both are monic of degree $p-1$ with the same roots mod p ,

$$P(x) \equiv x^{p-1} - 1 \pmod{p}.$$

Comparing coefficients, every intermediate symmetric sum vanishes mod p :

$$S_k \equiv 0 \pmod{p}, \quad 1 \leq k \leq p-2.$$

In particular $p \mid S_{p-2}$, hence $p \mid A$.

Divisibility by p^2 . Evaluate P at $x = p$:

$$P(p) = (p-1)(p-2)\cdots(1) = (p-1)!.$$

Expanding algebraically,

$$P(p) = p^{p-1} - S_1p^{p-2} + S_2p^{p-3} - \cdots - S_{p-2}p + S_{p-1},$$

where we used $(-1)^{p-1} = 1$ since p is odd. Since $S_{p-1} = (p-1)!$, cancelling and dividing by p :

$$0 = p^{p-2} - S_1p^{p-3} + \cdots + S_{p-3}p - S_{p-2},$$

that is,

$$S_{p-2} = p^{p-2} - S_1p^{p-3} + S_2p^{p-4} - \cdots + S_{p-3}p.$$

We analyse this modulo p^2 . For $p \geq 5$, the exponent $p-2 \geq 3$, so $p^{p-2} \equiv 0 \pmod{p^2}$. For each earlier term $S_j p^{p-2-j}$ with $1 \leq j \leq p-4$, the power of p is at least p^2 , so the term vanishes mod p^2 . Only the final term $S_{p-3} \cdot p$ survives:

$$S_{p-2} \equiv S_{p-3} p \pmod{p^2}.$$

By the result from **Divisibility by p** , $S_{p-3} \equiv 0 \pmod{p}$, so $S_{p-3} = mp$ for some integer m . Therefore

$$S_{p-2} \equiv (mp) \cdot p = mp^2 \equiv 0 \pmod{p^2}.$$

Hence $p^2 \mid S_{p-2}$, and since $(p-1)!$ is coprime to p , $p^2 \mid A$. ■